CSE320: Data Communications

Quiz 02 [Summer 2025]

Time: 20 minutes, Full Marks: 15

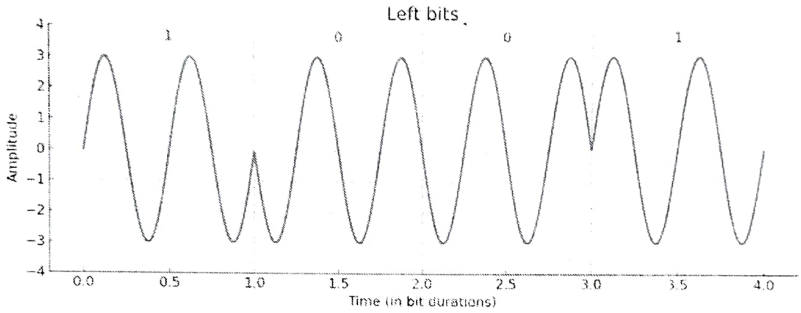
B

Name:		ID:		Section:	

1. We want to design an FSK scheme with a frequency shift of $\Delta f = 2$ **Hz** for a carrier signal with frequency $f_c = 4$ **Hz**, peak amplitude = **5V**, and phase = **90°**. Find out the frequencies for **1** and **0**. **[2]**

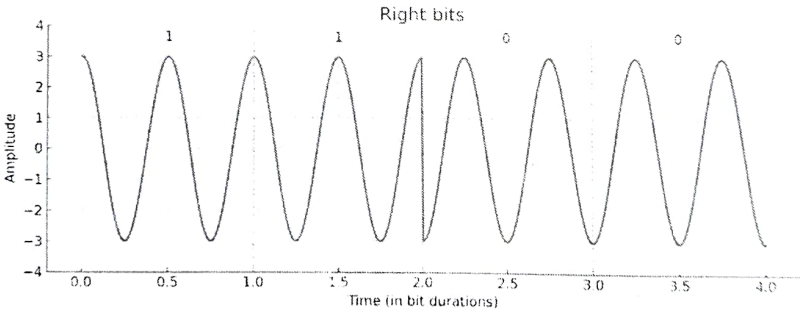
$f_1 = 4+2 = 6$ Hz, $f_0 = 4-2 = 2$ Hz

2. For the QPSK-modulation scheme given below, draw the **constellation diagram** and show the resultant **amplitudes** and **phases** clearly in a table. **[5]**



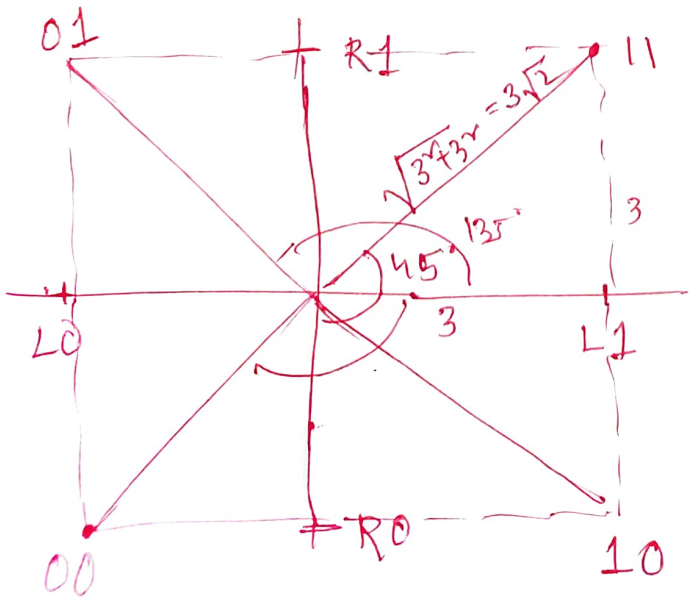
L1 : 0°

L0 : 180°

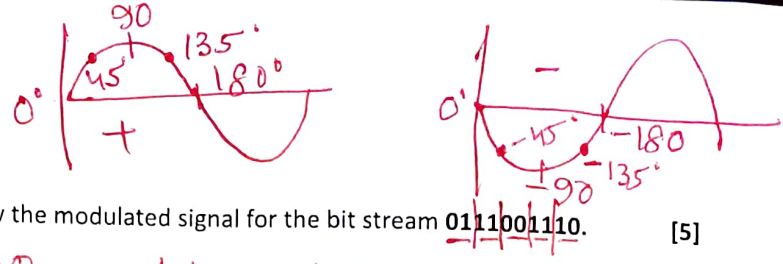


R1 : 90°

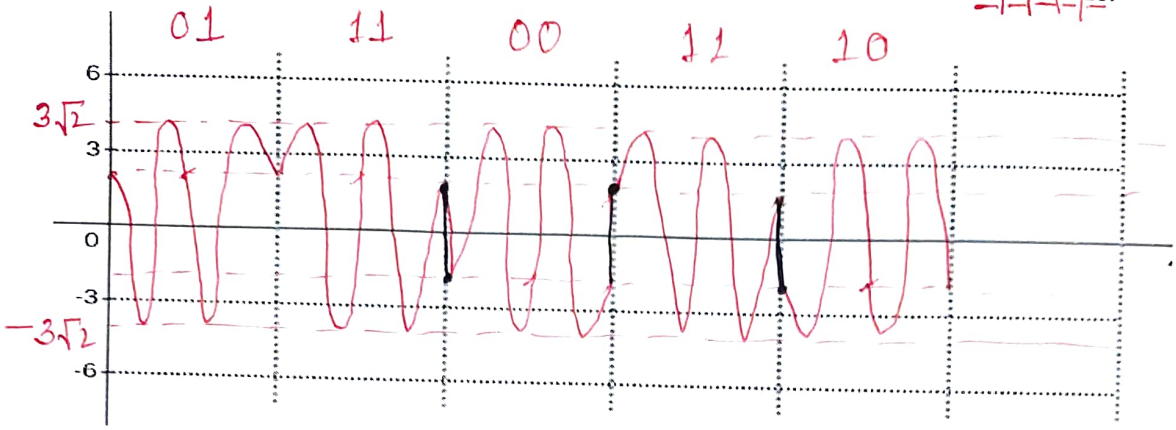
R0 : 270°



	amp	phase
00	$3\sqrt{2}$	−135°
01	$3\sqrt{2}$	135°
10	$3\sqrt{2}$	−45°
11	$3\sqrt{2}$	45°



3. Using the **QPSK scheme of question 2**, draw the modulated signal for the bit stream **01110011110.** **[5]**



4. Suppose we want to build an analog transmission system that can send **3-bits in the same baud.** However, we only have **2 voltage levels (3V and 5V)** and the standard **4 phases (0, 90, 180, 270)**. Show how we can send 3-bits in the same baud using the available voltages and phases only. **[3]**

000 ⟶ (3, 0°)
001 ⟶ (3, 90°)
010 ⟶ (3, 180°)
011 ⟶ (3, 270°)
100 ⟶ (5, 0°)
101 ⟶ (5, 90°)
110 ⟶ (5, 180°)
111 ⟶ (5, 270°)